LEARN
UNDERSTAND
REVISE
SUCCEED

CURRENT AFFAIRS
STATIC LINKAGES
EXAM FOCUS
VISUAL LEARNING

UPSC CURRENT AFFAIRS THROUGH INFOGRAPHICS

IMPORTANT ISSUES | KEY CONCEPTS | EXAM-ORIENTED INSIGHTS

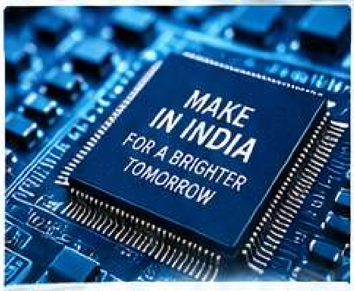
UPDATED FOR 2026

CONCISE | VISUAL | EXAM FOCUSED | RELIABLE | EASY REVISION

SCO BISHKEK DECLARATION
REGIONAL SECURITY & GLOBAL ORDER

IST RULES 2026
TIME FOR A DIGITAL INDIA

INDIA–BELGIUM RELATIONS
TRADE, TECHNOLOGY AND A STRONGER PARTNERSHIP



SEMICON INDIA 2026
SILICON TO SYSTEMS: **BUILDING** THE ECOSYSTEM

MARINE FISHERIES & INDIA'S BLUE ECONOMY
HEALTHY OCEANS PROSPEROUS PEOPLE

AI vs COPYRIGHT LAW
INNOVATION, CREATORS' RIGHTS AND THE ROAD AHEAD

AND MORE KEY DEVELOPMENTS SHAPING INDIA AND THE WORLD

A WELL-INFORMED ASPIRANT BUILDS A STRONGER INDIA

ECONOMY | INTERNATIONAL RELATIONS | SCIENCE & TECHNOLOGY | ENVIRONMENT | POLITY & GOVERNANCE | SOCIETY | DEVELOPMENT

FOR UPSC AND OTHER COMPETITIVE EXAMS

KNOWLEDGE TODAY A STRONGER TOMORROW

ISRO LAUNCHES
GSLV-F17 / EOS-05

Strengthening India's Earth Observation Capabilities

Launch Date	4 September 2026
Launch Vehicle	GSLV-F17 (19th flight of GSLV)
Launch Site	Satish Dhawan Space Centre, Sriharikota
Payload	EOS-05 (Earth Observation Satellite)
Mission Status	Successfully launched

इसरो isro

SPACE FOR A SAFER, STRONGER AND A BRIGHTER INDIA

EOS-05
An advanced Earth observation satellite to provide high-resolution images for a wide range of applications.

ABOUT EOS-05

Type Earth Observation Satellite

Objective High-resolution imaging of the Earth's surface

Applications
- Agriculture and crop monitoring
- Disaster management
- Forest and environmental monitoring
- Urban and infrastructure planning
- Cartography and resource mapping

Builds on Success of the EOS series (e.g., Cartosat, Resourcesat)

GSLV-F17: KEY FEATURES

- **Three-stage launch vehicle** Solid + Liquid + Cryogenic
- Equipped with an **indigenous cryogenic engine** (CE-7.5)
- **19th flight** of GSLV
- Designed to place heavier satellites in higher orbits
- Demonstrates India's maturity in cryogenic technology

WHY IT MATTERS?

- Enhances India's Earth observation capabilities
- Supports disaster preparedness and climate action
- Aids in sustainable development and resource management
- Strengthens India's self-reliance in space technology
- Contributes to national security and strategic planning

COMPARISON: PSLV vs GSLV vs LVM3

Feature	PSLV	GSLV (Mk II)	LVM3 (Mk III)
Full Form	Polar Satellite Launch Vehicle	Geosynchronous Satellite Launch Vehicle	Launch Vehicle Mark 3
Configuration	4-stage (S-L-S-L)	3-stage (S-L-C)	3-stage (S-L-C) with strap-ons
Payload Capacity	~1,750 kg to LEO	~2,500 kg to GTO	~4,000 kg to GTO ~8,000 kg to LEO
Typical Use	Small and medium satellites, Earth observation	Heavier satellites, communication satellites	Very heavy satellites, human spaceflight (Gaganyaan)
Key Highlights	Workhorse of ISRO, high reliability	Uses cryogenic engine (CE-7.5)	Most powerful Indian launch vehicle, enables big missions

WHAT IS A CRYOGENIC ENGINE?

- Uses liquid hydrogen (fuel) and liquid oxygen (oxidiser) at extremely low temperatures.
- Provides high thrust and efficiency.
- Essential for placing heavy satellites in higher orbits (GTO).
- Indigenous cryogenic engine (CE-7.5) marks a major technological milestone for India.

TYPES OF ORBITS

POLAR ORBIT (LEO)

- Low Earth Orbit (LEO) ~500–800 km
- Passes over the poles
- Useful for Earth observation, mapping, surveillance
- Examples: Cartosat, Resourcesat, EOS series

GEOSTATIONARY ORBIT (GEO)

- ~35,786 km above Earth
- Hovers over the equator
- Appears fixed over a point on Earth
- Used for communication, weather satellites
- Examples: INSAT, GSAT series

THE BIGGER PICTURE

- Strengthens India's position as a leading space power
- Supports Atmanirbhar Bharat in high-technology sectors
- Enables data-driven governance and development
- Contributes to global efforts on climate change, disaster resilience and sustainable future

INDIA REACHES HIGHER

VIKSIT BHARAT THROUGH SPACE

INDIA'S GDP GROWS 7.8% IN Q1 FY27

Strong domestic demand and record exports support growth

GS-III | ECONOMY

FOR A STRONGER INDIA

HIGHER GROWTH
STRONGER INDIA

Real GDP Growth (Q1 FY27)
7.8%
(Apr–Jun 2026)

India's Goods Exports (June 2026)
$44.24 billion
(Record high)

Key Driver
Strong exports, resilient domestic demand

Challenge
Lower-margin / labour-intensive exports under pressure

WHAT IS GDP?

Gross Domestic Product (GDP) is the total monetary value of all final goods and services produced within a country's borders in a given period of time.

- **Goods** (e.g., food, cars)
- **Services** (e.g., banking, IT)
- **Only final output is counted** (to avoid double counting)

GDP vs GVA

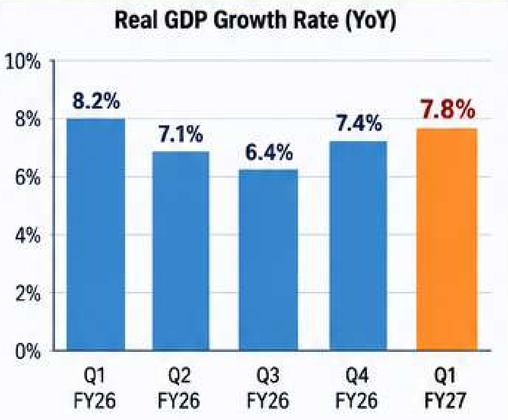
GDP (Gross Domestic Product)	• Value of all final goods and services • Includes indirect taxes minus subsidies
GVA (Gross Value Added)	• Value added at each stage of production • Does not include indirect taxes and subsidies

Relationship: GDP = GVA + Indirect Taxes – Subsidies

REAL vs NOMINAL GDP

Nominal GDP	• Measured at current market prices • Affected by inflation
Real GDP	• Measured at constant (base year) prices • Shows actual growth in physical output • Q1 FY27 growth (7.8%) is real GDP growth

TRENDS IN INDIA'S GDP GROWTH



HOW GDP IS CALCULATED? (Expenditure Method)

$$GDP = C + I + G + (X - M)$$

C	Private Final Consumption Expenditure (Households)
I	Gross Fixed Capital Formation (Investment)
G	Government Final Consumption Expenditure
X	Exports of Goods and Services
M	Imports of Goods and Services (Net exports = X – M)

EXPORTS AND GDP

- India's goods exports rose to $44.24 billion in June 2026 (record high).
- Higher exports increase net exports (X – M), directly adding to GDP.
- Key sectors: engineering goods, petroleum products, electronics, pharmaceuticals, chemicals.
- However, lower-margin and labour-intensive exports face global headwinds (rising competition, trade barriers, slower demand).

WHY GROWTH MATTERS?

- More jobs and higher incomes
- Supports manufacturing and services expansion
- Increases government revenue for welfare and infrastructure
- Helps achieve Viksit Bharat @ 2047

FACTORS SUPPORTING Q1 FY27 GROWTH

- Strong goods exports (record $44.24 billion in June)
- Resilient domestic consumption
- Continued investment activity
- Government capital expenditure
- Stable macroeconomic fundamentals

KEY CHALLENGES AHEAD

- Global economic uncertainty and trade barriers
- Pressure on labour-intensive exports
- Commodity price volatility
- Need to sustain investment and job creation

KEY TAKEAWAYS

1. India's real GDP grew 7.8% in Q1 FY27.
2. Exports are a major growth driver, with goods exports at $44.24 billion in June.
3. Understand GDP vs GVA, real vs nominal GDP and the expenditure method.
4. High growth supports jobs, investments and inclusive development, but external risks remain.
5. Sustained reforms and global competitiveness are key for long-term growth.

A STRONG ECONOMY
A BRIGHTER TOMORROW

PEOPLE | PRODUCTIVITY | PROSPERITY

VIKSIT BHARAT @ 2047

GS-II | INTERNATIONAL RELATIONS

SHANGHAI COOPERATION ORGANISATION

SCO BISHKEK DECLARATION

Stronger Cooperation for a Stable, Secure and Prosperous Region

SCO SUMMIT 2026
BISHKEK, KYRGYZSTAN

Dialogue | Security | Development

KEY HIGHLIGHTS OF THE BISHKEK DECLARATION

- **Strong opposition to terrorism, separatism and extremism.**
- Rejection of **"double standards"** in counter-terrorism efforts.
- Emphasis on **dialogue and diplomacy** for resolving conflicts.
- Commitment to **regional stability, connectivity and** economic cooperation.

FROM SHANGHAI FIVE TO SCO

- **1996:** Shanghai Five formed by China, Russia, Kazakhstan, Kyrgyzstan and Tajikistan to build trust and reduce border tensions.
- **2001:** Uzbekistan joined and the grouping was formally transformed into the Shanghai Cooperation Organisation (SCO).
- **Gradual expansion** with new members, observers and dialogue partners.

ABOUT SCO

- **Established:** 2001 (evolved from Shanghai Five, 1996)
- **Objective:** Promote regional security, stability, economic cooperation and cultural ties.
- **Nature:** A Eurasian political, economic and security grouping.
- **Secretariat:** Beijing, China
- **Current Chair:** Kyrgyzstan (2026)
- **Next Chair:** Tajikistan (2027)
- **Working Mechanism:** Annual summits, Council of Heads of State, Council of Ministers, RATS, sectoral meetings.

MEMBERS, OBSERVERS & DIALOGUE PARTNERS

Members (10)

India, China, Russia, Pakistan, Kazakhstan, Kyrgyzstan, Tajikistan, Uzbekistan, Iran, Belarus

Observer States (2)

Mongolia, Afghanistan

Dialogue Partners (14)

Türkiye, Sri Lanka, Saudi Arabia, UAE, Egypt, Qatar, Bahrain, Kuwait, Maldives, Myanmar

(Others: Armenia, Azerbaijan, Cambodia, Nepal)

RATS – REGIONAL ANTI-TERRORIST STRUCTURE

- **Established:** 2004 (Tashkent, Uzbekistan)
- **Function:** Coordinate efforts against terrorism, separatism and extremism.
- **Activities:** Intelligence sharing, capacity building, joint exercises, database of terrorist organisations.
- **Significance:** Institutional mechanism to address common security threats in the region.

INDIA AND SCO

- Joined as a full member in 2017 (along with Pakistan).
- Actively participates in counter-terrorism, connectivity, economic and cultural initiatives.
- At the Bishkek Summit, India reiterated dialogue and diplomacy for resolving conflicts and highlighted the need for a rules-based, multilateral order.
- Supports a united stance against terrorism and rejects double standards.
- Uses SCO as a platform to engage with Central Asian states and Eurasia.

INDIA'S CENTRAL ASIA POLICY

- Strengthen historical and cultural ties with Central Asian countries.
- Expand cooperation in trade, connectivity, energy, defence, critical minerals and education.
- **Key initiatives:** International North-South Transport Corridor (INSTC), Chabahar Port, Central Asia–India Dialogue, development partnerships.
- Important for energy security, access to resources and balancing regional powers.
- Part of India's broader "Extended Neighbourhood" and Act East–Connect Central Asia approach.

CHINA/RUSSIA FACTOR

China

- China seeks to use SCO to expand its influence and advance BRI.
- India remains cautious about China's strategic assertiveness.

Russia

- Russia views SCO as a key platform to maintain its influence in Eurasia.
- India continues to engage with Russia while balancing its ties with other partners.

INDIA'S STRATEGIC INTERESTS IN SCO

- Counter-terrorism cooperation.
- Access to Central Asian markets and resources.
- Connectivity (INSTC, Chabahar, regional corridors).
- Balancing China's influence.
- Energy security and critical minerals.
- Supporting a multipolar, rules-based international order.

PEACEFUL NEIGHBOURHOOD
PROSPEROUS REGION
STRONGER INDIA

COOPERATION TODAY
A MORE SECURE TOMORROW

INDIAN STANDARD TIME (IST) RULES 2026

A common, legally enforceable time reference for a digitally connected India

SYNCHRONISED TIME STRONGER INDIA

IST
UTC + 5:30
India's Official Time Reference

ONE TIME ONE INDIA | ACCURACY RELIABILITY | SECURITY RESILIENCE | ENABLING DEVELOPMENT

WHAT ARE THE IST RULES 2026?

- A new legal and technical framework to define, maintain and disseminate Indian Standard Time (IST).
- Provides a common, accurate and enforceable national time reference for all sectors.
- Ensures synchronisation of time across government, industry and citizens.
- Brings timekeeping under a robust regulatory and technical system.

KEY PROVISIONS

- Legally enforceable Indian Standard Time
- Authorises **CSIR–NPL** as the national timekeeper
- Standardised time dissemination across telecom, internet, broadcast and critical infrastructure
- Periodical calibration and audit mechanisms
- Defines responsibilities for government and private entities
- Measures for security, redundancy and resilience

WHO MAINTAINS IST?

CSIR-NPL
National Physical Laboratory, New Delhi

वैज्ञानिक सेवा राष्ट्र सेवा

- India's apex institute for timekeeping.
- Maintains IST using advanced atomic clocks.
- Generates and disseminates time signals through multiple channels.
- Works under the Council of Scientific and Industrial Research (CSIR), Ministry of Science & Technology.

CSIR-NPL is India's official source for Indian Standard Time.

HOW IS IST GENERATED?

ATOMIC CLOCKS
Ultra-precise cesium atomic clocks at CSIR-NPL measure time.

↓

UTC COMPARISON
IST is kept in sync with Coordinated Universal Time (UTC) with an offset of +5:30 hours.

↓

DISSEMINATION
Time signals are shared via telecom networks, internet (NTP), broadcast, and dedicated services to government and industry.

UNDERSTANDING THE TIME TERMS

UTC (Coordinated Universal Time)
- Global time standard maintained by BIPM.
- Based on atomic clocks.
- Not affected by time zones.

IST (Indian Standard Time)
- UTC + 5:30 hours (5 hours 30 minutes ahead).
- Single time zone for the entire country.
- No daylight saving time.

ATOMIC CLOCKS
- Use the natural vibration of atoms (usually cesium).
- Extremely accurate (error of less than a second in millions of years).
- Foundation for UTC and IST.

WHY SYNCHRONISED TIME MATTERS?

Digital Infrastructure
Essential for telecom, internet, 5G/6G, cloud services and data centres.

Cybersecurity
Prevents time spoofing; critical for encryption, digital signatures and secure transactions.

Financial Systems
Accurate time is vital for stock markets, banking and digital payments.

Transport & Logistics
Navigation, railways, aviation and shipping depend on precise time.

Power Grids
Synchronisation ensures stable and reliable electricity supply.

Governance & Public Services
Supports e-governance, emergency services and legal processes.

Space & Defence
Critical for satellite operations, navigation (GPS/NavIC) and secure communications.

LEGAL METROLOGY PERSPECTIVE

- Time measurement falls under the Legal Metrology Act, 2009.
- The IST Rules 2026 provide the legal basis for enforcement, compliance and penalties in case of tampering or misuse.
- Ensures uniformity across industries and service providers.

KEY TAKEAWAYS

1. IST Rules 2026 provide a common and enforceable national time reference.
2. CSIR-NPL maintains IST using atomic clocks.
3. IST = UTC + 5:30 (no daylight saving time).
4. Accurate time is critical for cybersecurity, finance, transport, power and digital India.
5. Strengthens trust, security and resilience in a connected economy.

THE BIGGER PICTURE

- A synchronised India
- A secure digital future
- Stronger economy
- Reliable infrastructure
- Greater national security

ACCURATE TIME CONNECTS PEOPLE POWERS PROGRESS

SAME TIME A STRONGER TOMORROW

INDIA–BELGIUM RELATIONS UPGRADED

Stronger Partnerships for a Shared Prosperity

DEMOCRACIES PARTNERS FOR A BETTER TOMORROW

Grand Place, Brussels (Belgium)

- STRONGER PEOPLE-TO-PEOPLE TIES
- CLEANER GREENER FUTURE
- INNOVATION & TECHNOLOGY
- SHARED VALUES A STRONGER WORLD

KEY ANNOUNCEMENTS

- India and Belgium agreed to deepen cooperation in trade, defence, clean energy, critical minerals, semiconductors, research and talent mobility.
- **Investment Fast-Track Mechanism** to facilitate and accelerate investments.
- **Goal to double bilateral trade within five years.**
- Enhanced collaboration in education, research, innovation and talent mobility.
- Commitment to a long-term, strategic partnership based on shared democratic values.

AREAS OF COOPERATION

Trade & Investment	Defence & Security	Clean Energy & Climate Action
Critical Minerals	Semiconductors & Advanced Manufacturing	Research, Innovation & Technology
Education & Talent Mobility	People-to-People Ties	Sustainable Development

WHY BELGIUM MATTERS?

Strategic location
Gateway to the European Union, hosts EU institutions (EU Commission and EU Council).

Economic strength
Highly developed economy with strengths in chemicals, pharmaceuticals, engineering, logistics and high-tech industries.

Innovation hub
Home to world-class research institutions, universities and R&D ecosystem.

Key role in global supply chains
Important player in semiconductors, pharmaceuticals and critical raw materials.

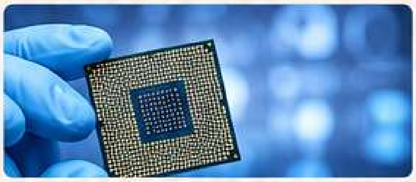
Influence in global governance
Active in multilateral forums, supports a rules-based international order.

INDIA–EU RELATIONS CONTEXT

- Belgium is an important partner within the European Union.
- India–EU Strategic Partnership covers trade, connectivity, climate change, clean energy, digital technology and people-to-people ties.
- Belgium often supports closer India–EU economic and political engagement.
- Forthcoming India–EU Free Trade Agreement (FTA) will further strengthen ties.

Stronger India–Belgium ties contribute to a deeper and broader India–EU partnership.

SEMICONDUCTORS



- Belgium is a key player in the global semiconductor supply chain (e.g., advanced materials, equipment, research).
- Cooperation can support India's semiconductor mission through technology, R&D and skilled talent.
- Reduces dependence on limited geographies and strengthens supply-chain resilience.
- Opens opportunities for joint research, manufacturing and innovation.

CRITICAL MINERALS

- Belgium plays an important role in trading and processing critical minerals.
- Cooperation ensures secure, diversified and sustainable supply chains for India's clean energy transition.
- Helps India's green hydrogen, electric vehicles, renewable energy and high-tech manufacturing sectors.
- Aligns with shared goal of building resilient and responsible mineral supply chains.

BILATERAL TRADE

Current trade (2024-25)
~ US$ 13.5 billion (approx.)

Target
Double trade within 5 years

Major items of trade

India's exports
Pharmaceuticals, chemicals, textiles, machinery, agri-products, IT services

India's imports
Diamonds, machinery, chemicals, electronic goods, metals and high-tech products

KEY TAKEAWAYS

1. Upgradation reflects growing trust and convergence of interests.
2. Opens new opportunities in trade, defence, clean energy, critical minerals and semiconductors.
3. Supports India's economic growth, technological advancement and green transition.
4. Strengthens India–EU partnership and India's global standing.
5. A win-win partnership for a stable, prosperous and sustainable future.

Atomium, Brussels
(Symbol of innovation and progress)

PEOPLE | PARTNERSHIP | PROGRESS

STRONGER TODAY
BRIGHTER TOMORROW

SEMICON INDIA 2026

TECHNOLOGY
INNOVATION
ATMANIRBHAR BHARAT

Silicon to Systems: Building the Ecosystem

India's platform to power a self-reliant and globally connected semiconductor ecosystem

Dates	Venue	Edition	Theme
17 – 19 September 2026	New Delhi (India)	5th Edition	Silicon to Systems: Building the Ecosystem

ABOUT SEMICON INDIA

- Flagship event bringing together government, industry, investors, innovators and global partners.
- Showcases opportunities across the semiconductor value chain.
- Platform for policy dialogue, partnerships, technology showcase and investment promotion.
- Positions India as a key player in the global semiconductor ecosystem.

KEY FOCUS AREAS (2026)

- Manufacturing and supply chains
- ATMP/OSAT and design ecosystem
- Emerging technologies (AI, EV, IoT, 5G)
- Global partnerships and investments
- Startup innovation and talent development
- Resilient and secure supply chains
- Sustainability and green semiconductors

ORGANISED BY

Ministry of Electronics and Information Technology
Government of India

सत्यमेव जयते

semicon india
BUILDING A RESILIENT SEMICONDUCTOR ECOSYSTEM

Towards a Viksit Bharat through Semiconductors

INDIA SEMICONDUCTOR MISSION (ISM)

INDIA SEMICONDUCTOR MISSION
Catalyzing India's Semiconductor Ecosystem

- Launched in 2021 under MeitY.
- Objective: Develop a vibrant semiconductor and display manufacturing ecosystem in India.
- Financial support for fabs, ATMP/OSAT units, design infrastructure, and R&D.
- Focus on creating a self-reliant, innovation-driven and globally competitive semiconductor sector.
- Total outlay: ~₹76,000 crore.

SEMICON 2.0 (NEXT PHASE)

- Builds on the momentum of ISM.
- Focus on larger, more advanced fabs.
- Greater support for design, R&D and compound semiconductors (SiC, GaN).
- Incentives for equipment & materials manufacturing.
- Skilling and talent development at scale.
- Stronger integration with global supply chains.
- Aims to position India as a global semiconductor hub.

KEY SEGMENTS OF THE VALUE CHAIN

Fab (Wafer Fabrication) →	ATMP/OSAT (Assembly, Testing, Marking & Packaging) →	Design
Manufactures semiconductor wafers (front-end processes).	Packages and tests chips (back-end processes).	Chip design, IP cores and EDA tools.

Complete ecosystem: Materials + Equipment + Design + Manufacturing + Packaging + Testing + End-use industries

SUPPLY-CHAIN SECURITY

- Reduces dependence on a single country for critical chips.
- Diversifies sources of raw materials, equipment and technology.
- Builds domestic manufacturing and trusted supply chains.
- Enhances resilience against geopolitical shocks and disruptions.

CHINA–US COMPETITION: IMPLICATIONS

China

- Dominates global chip manufacturing (high share in mature nodes).
- Strong control over critical minerals and supply chains.
- US export restrictions on advanced chips and equipment.
- India seeks to reduce over-dependence on China.

United States

- Leads in advanced chip design, technology and equipment.
- Imposes export controls to limit China's access to high-end chips.
- Seeks trusted partners to diversify supply chains ("China+1").
- India is seen as a key partner in building resilient supply chains.

WHY IT MATTERS FOR INDIA?

- Supports Atmanirbhar Bharat and economic growth.
- Generates high-skilled jobs.
- Enables domestic manufacturing in strategic sectors (electronics, defence, automotive, telecom, AI).
- Strengthens technological sovereignty and national security.
- Positions India in the evolving global semiconductor value chain.

FROM CHIPS TO A STRONGER INDIA

SELF-RELIANCE | INNOVATION | GLOBAL PARTNERSHIP | SUSTAINABLE GROWTH

SEMICONDUCTORS POWER PEOPLE PROSPERITY

A STRONGER TECH INDIA
A BRIGHTER TOMORROW

SEMICON INDIA 2026 | IDEAS | INVESTMENTS | INNOVATION | A STRONGER, SELF-RELIANT INDIA

VIKSIT BHARAT 2047

MARINE FISHERIES & INDIA'S BLUE ECONOMY

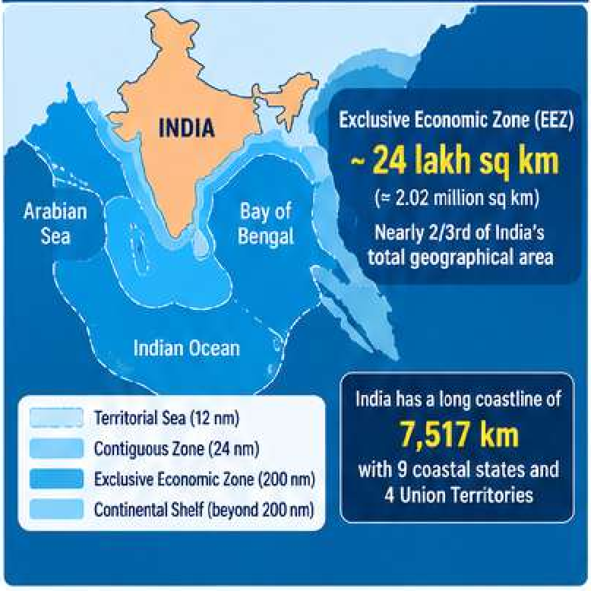
Healthy Oceans, Prosperous People, Stronger India

CLEAN OCEANS
THRIVING COASTS
VIKSIT BHARAT

FOOD NUTRITION | MILLIONS OF LIVELIHOODS | EXPORT EARNINGS | SUSTAINABLE OCEAN GROWTH

BLUE ECONOMY
STRONG OCEANS
STRONGER INDIA

INDIA'S MARINE RESOURCES



Exclusive Economic Zone (EEZ)
~ 24 lakh sq km
(≈ 2.02 million sq km)
Nearly 2/3rd of India's total geographical area

India has a long coastline of
7,517 km
with 9 coastal states and 4 Union Territories

MARINE FISHERIES AT A GLANCE

4.3 million tonnes
Marine fish production (2023–24, approx.)

~ 4 million
People depend on marine fisheries (fishers, fish farmers, workers, allied activities)

~ ₹60,000+ crore
Export earnings from marine products (2023–24)

Key to nutrition
Provides high-quality protein, essential omega-3 fatty acids and micronutrients

Major species
Tuna, shrimp, sardine, mackerel, seer fish, pomfret, squid, cuttlefish, lobster, crab etc.

WHAT IS BLUE ECONOMY?

Sustainable use of ocean resources for economic growth, improved livelihoods and ocean ecosystem health.

- Fisheries & Aquaculture
- Maritime Transport & Ports
- Offshore Energy (oil, gas, renewables)
- Coastal Tourism
- Marine Biodiversity
- Blue Technology & Ocean Innovation
- Seabed Minerals (Polymetallic nodules, polymetallic sulphides, etc.)

Blue Economy = Economic Growth + Healthy Oceans + Resilient Coastal Communities

PM MATSYA SAMPADA YOJANA (PMMSY)

(2020–21 to 2024–25)

Objective: Sustainable and responsible development of fisheries sector.

Total Investment: ~ ₹20,050 crore
(Central share ~ ₹9,407 crore)

Key Components:

- Enhancing fish production and productivity
- Infrastructure (fishing harbours, landing centres, cold chain)
- Fisheries value chain and marketing
- Aquaculture development
- Fishers' welfare and social security
- Monitoring, control and surveillance (against IUU fishing)

SUSTAINABLE FISHING & CONSERVATION

Prevent overfishing
Follow science-based catch limits and seasonal bans (e.g., breeding season).

Reduce bycatch
Use selective and eco-friendly fishing gear.

Protect marine ecosystems
Conserve coral reefs, mangroves, seagrass and breeding grounds.

Combat IUU fishing
Strengthen monitoring, control and surveillance (MCS) using technology (VMS, AIS, drones).

Promote responsible aquaculture
Adopt low-impact, climate-resilient and sustainable practices.

UNCLOS & INDIA

- UNCLOS (1982) is the legal framework governing oceans and their resources.
- Defines maritime zones:
  - Territorial Sea – up to 12 nm
  - Contiguous Zone – up to 24 nm
  - **Exclusive Economic Zone (EEZ) – up to 200 nm**
  - Continental Shelf – beyond 200 nm (up to 350 nm in some cases)
- Gives sovereign rights over resource exploration and exploitation in the EEZ and continental shelf.
- India is a party to UNCLOS and actively participates in its institutions (e.g., ISA – International Seabed Authority).

UNCLOS enables India to harness its ocean wealth while ensuring responsible and sustainable use.

CHALLENGES

- Overfishing and declining fish stocks
- Climate change (ocean warming, acidification)
- Marine pollution (plastic, oil spills, ghost nets)
- IUU fishing by foreign vessels
- Habitat loss (mangroves, coral reefs)
- Livelihood vulnerability of coastal communities
- Need for better data, research and enforcement

OPPORTUNITIES

- Rising global demand for seafood
- Value addition and export diversification
- Deep-sea fishing and under-exploited resources (e.g., tuna)
- Mariculture and seaweed cultivation
- Ocean-based renewable energy (offshore wind, tidal)
- Blue biotechnology and pharmaceuticals
- Coastal and island tourism
- Employment generation and women's participation

WAY FORWARD

- Implement science-based fisheries management
- Strengthen MCS and curb IUU fishing
- Invest in sustainable aquaculture and eco-friendly infrastructure
- Empower fishers with technology, credit and social security
- Enhance research in marine biodiversity and climate resilience
- Regional and global cooperation for healthy oceans
- Mainstream Blue Economy in national growth strategy

"Healthy Oceans Today for Prosperous Generations Tomorrow"

VIKSIT BHARAT 2047

SUSTAINABLE RESOURCES | RESILIENT COMMUNITIES | THRIVING ECONOMY | CLEANER OCEANS

AI VS COPYRIGHT LAW

Innovation, Intellectual Property and a Fair Future

How do we balance the promise of generative AI with the rights of creators?

CREATIVITY FUELS HUMAN PROGRESS. LET IT THRIVE IN THE AI ERA.

GS-II / GS-III
UPSC CURRENT AFFAIRS

THE ISSUE

- Generative AI models are trained on massive amounts of data, often including copyrighted material (text, images, music, etc.).
- Creators and publishers allege unauthorised use of their works.
- AI companies argue it is necessary for innovation and may be covered under exceptions like fair dealing (fair use).
- The debate is about how to balance creators' rights with technological progress.

KEY STAKEHOLDERS

Creators
(authors, artists, musicians, filmmakers, publishers)
Seek recognition, consent and fair compensation.

AI companies
Need large datasets to train models; argue for flexibility and legal clarity.

Users / Public
Access to innovative and affordable AI products.

Governments & Regulators
Balance innovation, IP rights and public interest.

RELEVANT LEGAL FRAMEWORK

Copyright Act, 1957 (India)

- Protects original literary, dramatic, musical and artistic works, etc.
- Rights include reproduction, communication to the public, adaptation and translation.
- Duration: Generally life of author + 60 years.

Key Provisions / Concepts

- **Section 14** – exclusive rights of copyright owner.
- **Section 51** – what constitutes infringement.
- **Section 52** – exceptions (fair dealing for research, private use, criticism, review, reporting of current events, etc.).
- **Section 57** – moral rights (right to attribution, integrity).

FAIR DEALING IN THE AI CONTEXT

AI training may be argued as:

- "Fair dealing" for research or non-commercial purposes.
- "Transformative use" (creates new outputs, not a substitute).

- Concerns:
  - Large-scale, commercial use may go beyond fair dealing.
  - Exact reproduction or close outputs can harm the market for original works.

Whether AI training is 'fair dealing' is a key legal question, and courts worldwide are examining it.

KEY CONCERNS

- Unauthorised use of copyrighted works for training data.
- Loss of income for creators and industries (music, publishing, media).
- Risk of plagiarism and generation of similar/competing works.
- Lack of transparency about training data.
- Bias and misuse of copyrighted or sensitive content.
- Impact on cultural diversity and traditional knowledge.

GLOBAL DEVELOPMENTS

United States
- Courts examining 'fair use' in AI training (e.g., lawsuits by authors, news organisations, artists).
- No specific AI copyright law yet.

European Union
- EU AI Act (2024) requires transparency about training data and respect for copyright.
- Text and Data Mining (TDM) exceptions with opt-out for rightsholders.

United Kingdom
- Consultations on copyright and AI.
- Proposes a more flexible regime with transparency requirements.

China
- Guidelines recognise copyright protection, with some allowances for training under specific conditions.

INDIA'S APPROACH

- No specific AI law yet; existing Copyright Act, 1957 applies.
- Government and expert bodies are examining the issue (consultations, committee discussions).
- Focus areas:
  - Need for transparency in training data.
  - Possible licensing or collective management systems.
  - Protection of creators' rights while enabling innovation and AI for public good.
- Also linked to India's digital economy, AI strategy and creative industry growth.

India needs a balanced, future-ready framework that protects creators, promotes innovation and leverages AI for inclusive growth.

POSSIBLE SOLUTIONS

Licensing models
Structured access to copyrighted content.

Collective management organisations
Pooling of rights and revenue sharing.

Transparency requirements
Disclosure of training data sources.

Compensation mechanisms
Revenue sharing or data compensation funds.

Clear guidelines on fair dealing
Define permissible uses for AI training.

Promote open datasets
Publicly available, licensed content.

IMPLICATIONS FOR UPSC

GS-II (Polity & Governance)
- Intellectual property rights
- Regulation of emerging technologies
- Role of government in balancing innovation and rights

GS-III (Science & Technology)
- AI and its applications
- Impact on economy and creative industries
- Ethical and legal challenges in technology

Essay / Ethics / Interdisciplinary
- Innovation vs. rights
- Technology and human values
- Inclusive growth in the digital age

KEY TAKEAWAYS

1. Generative AI has intensified the debate on copyright, with no easy answers.
2. Copyright Act, 1957 provides protection, but its application to AI training is evolving.
3. Global approaches vary — from flexible exceptions to stricter transparency norms.
4. India needs a balanced framework that ensures creator compensation, encourages innovation and aligns with global best practices.
5. The issue has far-reaching implications for the creative economy, digital rights and the future of AI governance.

SUPPORT CREATIVITY. ENABLE INNOVATION. BUILD A FAIR AI FUTURE.

PEOPLE IDEAS PROGRESS